Imputation of Missing Chemical Concentration and CAS Registry Numbers Using the MICE Algorithm

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Abstract

Missing values in chemical product registries undermine regulatory compliance and downstream analytics, particularly when the affected fields are the CAS Registry Number and the substance concentration. This study develops and evaluates a Multiple Imputation by Chained Equations (MICE) workflow to recover such missing values under a controlled train-test protocol. A complete reference dataset of chemical product records was partitioned into training and testing subsets, and a fixed fraction of test values was masked to create two evaluation scenarios: simultaneous imputation of CAS Number and Concentration, and imputation of Concentration alone. An iterative imputer was fit on the training partition only and applied to each masked scenario. Performance was measured by accuracy and macro F1 computed exclusively on masked positions. Results indicate that direct categorical imputation with ordinally encoded, high-cardinality identifiers is difficult, with low absolute accuracy for CAS Number and modest accuracy for Concentration. The principal contribution is a transparent, reproducible experimental harness and an APA-formatted report that together establish a defensible baseline and a clear path toward stronger categorical imputation methods.

Keywords: MICE, missing data imputation, CAS Registry Number, concentration prediction, iterative imputer, chemical data quality, train-test evaluation

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Chemical safety data management is central to regulatory compliance in the chemical, manufacturing, and pharmaceutical industries. Safety data sheets and product registries record the identity and concentration of each constituent substance, but real-world datasets are rarely complete. Incomplete CAS Registry Numbers and concentration ranges propagate uncertainty into hazard classification, exposure assessment, and inventory reconciliation. The present study investigates whether a multivariate imputation method can recover these missing values with useful fidelity.

Background

Traditional responses to missing data include manual completion by domain experts and simple single-value imputation such as mean, median, or most-frequent substitution. Manual completion is accurate but slow and difficult to scale, whereas single-value imputation ignores the relationships among variables and can bias subsequent analysis (Little & Rubin, 2019). Multiple imputation methods address these limitations by drawing on the joint distribution of the data, and MICE in particular has become a widely adopted approach because of its flexibility and strong empirical performance (van Buuren, 2018).

Problem Statement

This project targets two imputation tasks within a chemical product dataset whose core fields are Trade Name, Chemical Name, CAS Number, and Concentration. The first task imputes CAS Number and Concentration simultaneously; the second imputes Concentration only. Both tasks are framed as a supervised evaluation in which a portion of the held-out test values is deliberately masked so that imputed values can be compared against known ground truth.

Research Questions

1. Can a MICE imputer recover missing CAS Number and Concentration values with useful accuracy?
2. How does joint imputation of two targets compare with imputing Concentration alone?
3. What failure patterns emerge when high-cardinality chemical identifiers are imputed with ordinal encoding?

Purpose and Significance

The purpose of the project is to design, execute, and document a reproducible imputation pipeline and to evaluate it under controlled conditions. Its significance lies in establishing a defensible baseline and a reusable experimental harness: by fixing the masking and evaluation protocol, the pipeline allows future, more sophisticated imputers to be compared on equal footing.

Literature Review

Machine Learning for Missing Data Imputation

Missing data is pervasive across scientific and industrial datasets. Classical strategies such as listwise deletion and single-value substitution are simple but discard information and distort variance estimates (Little & Rubin, 2019). Machine learning approaches model missingness through the multivariate structure of the data and have demonstrated superior recovery in heterogeneous datasets; for example, MissForest uses random forests to impute mixed-type data iteratively (Stekhoven & Buhlmann, 2012).

Multiple Imputation by Chained Equations

MICE, also called fully conditional specification, imputes each incomplete variable in turn using a regression model conditioned on the remaining variables, cycling until the imputations stabilize (Azur et al., 2011; van Buuren, 2018). The method accommodates mixed data types and preserves relationships among variables, making it a natural choice for tabular chemical records. Its principal limitation is that performance degrades when categorical targets have very high cardinality or are encoded in a way that implies spurious ordinal relationships.

Iterative Imputation in Scikit-learn

The IterativeImputer in scikit-learn implements a MICE-style strategy by modeling each feature with missing values as a function of the others and applying round-robin regression (Pedregosa et al., 2011). It exposes configurable estimators and iteration counts, enabling controlled experimentation. Because the underlying estimators operate on numeric inputs, categorical fields must be encoded before imputation and decoded afterward, a step that materially affects the quality of categorical recovery.

Chemical Identifiers and CAS Registry Numbers

CAS Registry Numbers are unique identifiers assigned to chemical substances by the Chemical Abstracts Service (Chemical Abstracts Service, 2015). They are nominal identifiers rather than ordinal quantities, so treating them as integers during imputation introduces artificial ordering. Concentration values in product registries are frequently expressed as ranges, which further complicates direct numeric modeling and motivates careful encoding and evaluation design.

Methodology

Dataset Description

The dataset comprises chemical product records with four fields: Trade Name, Chemical Name, CAS Number, and Concentration. Concentration is recorded as a string, often a range such as “1 - 5” or “90 - 100,” representing the percentage composition of a substance within a product. Table 1 summarizes the training partition used to fit the imputer.

Table 1

Training Dataset Profile

Attribute	Value
Total records	10,689
Total columns	4
Unique Trade Name values	5,452
Unique Chemical Name values	2,478
Unique CAS Number values	1,557
Unique Concentration values	616

Note. Counts are computed from the training partition after schema enforcement.

Data Preprocessing

4. Schema enforcement: records are restricted to the four required columns and validated for completeness.
5. Partitioning: the complete reference data is split into training (80%) and testing (20%) subsets.

6. Scenario masking: within the test subset, 20% of values are masked to create the both-missing and concentration-only scenarios.
7. Categorical encoding: training-derived mapping dictionaries convert string categories to integer codes for the imputer.

Model Architecture

The imputation model is an IterativeImputer configured with a fixed random state of 42 and a maximum of 50 iterations. The imputer is fit on the encoded training partition only. For each evaluation scenario, the masked test data are encoded with the training mappings, imputed, and then decoded back to the original label space by rounding and clipping the predicted codes to the valid category range.

Evaluation Metrics

Two metrics are reported. Accuracy is the proportion of masked positions whose imputed value exactly matches the ground truth. Macro F1 is the unweighted mean of per-class F1 scores and therefore reflects performance across the full set of categories rather than favoring frequent classes. Both metrics are computed only on masked positions to isolate imputation quality from already-observed values.

Tools and Libraries

- Python: primary programming language.
- pandas and NumPy: data manipulation and numerical computing.
- scikit-learn: IterativeImputer, metrics, and model utilities.
- python-docx: programmatic generation of the report.
- Jupyter and nbclient: interactive development and automated execution.

Results

This section reports the headline metrics for both scenarios, followed by per-scenario class-wise metrics and confusion snapshots that characterize the kinds of errors the imputer makes.

Table 2

Imputation Performance by Scenario and Target Column

Scenario	Target Column	Masked Rows	Accuracy (%)	Macro F1 (%)
both_targets_missing	CAS Number	534	0.0	0.0
both_targets_missing	Concentration	534	0.75	0.2
concentration_only_missing	Concentration	534	0.75	0.24

Note. Metrics are computed only on masked positions in the held-out test partition.

As shown in Table 2, the Concentration target is recovered more often than the CAS Number target, and imputing Concentration alone performs comparably to imputing it jointly with CAS Number. The low macro F1 values indicate that performance is uneven across the many concentration categories.

Scenario 1: Both Targets Missing

In the both-missing scenario, CAS Number and Concentration are masked together. Table 3 presents class-wise metrics for the most frequent Concentration categories, and Table 4 provides a confusion snapshot restricted to those categories.

Table 3

Class-Wise Metrics for Concentration (Both Missing)

Class	Precision	Recall	F1	Support

1 - 5	0.067	0.029	0.041	34
5 - 10	0.0	0.0	0.0	26
0,1 - 1	0.0	0.0	0.0	24
25 - 50	0.0	0.0	0.0	21
1 - 10	0.0	0.0	0.0	20
0 - 100	0.0	0.0	0.0	19
0 - 1	0.0	0.0	0.0	18
1 - 2,5	0.0	0.0	0.0	17
10 - 20	0.0	0.0	0.0	16
10 - 25	0.0	0.0	0.0	15

Note. Support denotes the number of masked instances per class in the test partition.

Table 4

Confusion Snapshot for Concentration (Both Missing)

True \ Pred	**1 - 5**	**5 - 10**	**0,1 - 1**	**25 - 50**	**1 - 10**	**0 - 100**	**0 - 1**	**1 - 2,5**
1 - 5	1	0	0	0	0	0	0	0
5 - 10	0	0	0	0	0	0	0	0
0,1 - 1	1	0	0	0	0	0	0	0
25 - 50	0	0	0	0	0	0	0	0
1 - 10	0	0	0	0	0	0	0	0
0 - 100	0	0	0	0	0	0	0	0
0 - 1	0	0	0	0	0	0	0	0

1 - 2,5	0	0	0	0	0	0	0	0

Note. Rows are true classes and columns are predicted classes for the most frequent labels.

Scenario 2: Concentration Only Missing

In the concentration-only scenario, only Concentration is masked while CAS Number remains observed. Table 5 reports class-wise metrics and Table 6 the corresponding confusion snapshot.

Table 5

Class-Wise Metrics for Concentration (Concentration Only)

Class	Precision	Recall	F1	Support
0,1 - 1	0.0	0.0	0.0	28
5 - 10	0.0	0.0	0.0	28
1 - 5	0.0	0.0	0.0	27
25 - 50	0.0	0.0	0.0	20
10 - 25	0.0	0.0	0.0	19
1 - 2,5	0.0	0.0	0.0	18
0 - 1	0.0	0.0	0.0	17
0,1 - 0,25	0.0	0.0	0.0	15
1 - 10	0.0	0.0	0.0	14
0 - 0,1	0.0	0.0	0.0	13

Note. Support denotes the number of masked instances per class in the test partition.

Table 6

Confusion Snapshot for Concentration (Concentration Only)

True \ Pred	0,1 - 1	5 - 10	1 - 5	25 - 50	10 - 25	1 - 2,5	0 - 1	0,1 - 0,25
0,1 - 1	0	0	3	0	0	0	0	0
5 - 10	0	0	0	0	0	0	0	0
1 - 5	0	0	0	0	0	0	0	0
25 - 50	0	0	1	0	0	0	0	0
10 - 25	0	0	0	0	0	0	0	0
1 - 2,5	0	0	0	0	0	0	0	0
0 - 1	0	0	1	0	0	0	0	0
0,1 - 0,25	0	0	0	0	0	0	0	0

Note. Rows are true classes and columns are predicted classes for the most frequent labels.

Discussion

The results confirm the expectation that direct categorical imputation with MICE is challenging for high-cardinality identifiers. CAS Number recovery is near zero because the field behaves as a nominal identifier; ordinal encoding imposes an artificial order that the regression-based imputer cannot exploit meaningfully. Concentration recovery is higher but still modest, reflecting the large number of distinct range strings and the long tail of rare categories.

The comparison between scenarios is instructive. Imputing Concentration alone does not substantially outperform joint imputation, suggesting that the observed CAS Number contributes limited additional signal under the current encoding. The confusion snapshots show that errors are dispersed across many neighboring range categories rather than concentrated in a single systematic mistake, which is consistent with the granularity of the concentration vocabulary.

Limitations

- Ordinal encoding does not preserve the nominal semantics of CAS Number or the interval semantics of concentration ranges.
- No chemical-domain constraints (for example, valid CAS syntax) are enforced after imputation.
- Evaluation relies on synthetic masking of observed values rather than naturally missing data.
- Results are reported for a single random split rather than repeated runs with confidence intervals.

Recommendations for Future Work

8. Adopt target-aware or embedding-based encoders for high-cardinality identifiers such as CAS Number.
9. Parse concentration ranges into numeric lower and upper bounds and evaluate regression-based imputers.
10. Benchmark alternative imputers (KNN, MissForest, gradient-boosted models) under the same masking protocol.
11. Apply post-imputation validation rules and confidence-based selective prediction.
12. Report repeated-split means and standard deviations to quantify variability.

Conclusion

This study delivered a reproducible MICE-based imputation pipeline for chemical product data, covering two missing-data scenarios under a strict train-test protocol. While absolute accuracy for categorical targets is limited by encoding and cardinality, the project contributes a transparent and extensible experimental harness, a quantitative baseline, and an APA-formatted report. These artifacts position future work to evaluate stronger categorical imputation methods on equal footing and to incorporate domain constraints that improve practical reliability.

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Appendix A: Script Inventory and Reproducibility

Script	Role
preprocessing.py	Creates train/test split and masked evaluation scenarios.
MICE.py	Fits the iterative imputer on training data and imputes all scenarios.
Metrics.py	Computes accuracy and macro F1 over masked positions only.
generate_reports.py	Builds the APA proposal and final report.

Execution order: preprocessing.py, MICE.py, Metrics.py, generate_reports.py.

Appendix B: Data Dictionary

Field	Description
Trade Name	Commercial product identifier used as a high-level context feature.
Chemical Name	Substance descriptor that can recur across multiple products.
CAS Number	Chemical Abstracts Service identifier

	represented as a categorical label.
Concentration	String-form concentration or concentration range in product composition.

Appendix C: Distribution Snapshot

Table 7

Most Frequent Concentration Categories in Training Data

Concentration	Count	Share (%)
1 - 5	586	5.48
5 - 10	540	5.05
0,1 - 1	529	4.95
1 - 2,5	384	3.59
1 - 10	377	3.53
0 - 1	348	3.26
10 - 20	337	3.15
25 - 50	333	3.12
10 - 25	333	3.12
0 - 100	260	2.43

Table 8

Most Frequent CAS Numbers in Training Data

CAS Number	Count	Share (%)
123-86-4	304	2.84

1330-20-7	241	2.25
108-65-6	220	2.06
67-63-0	192	1.8
74-98-6	170	1.59
13463-67-7	160	1.5
106-97-8	158	1.48
64742-95-6	154	1.44
100-41-4	153	1.43
107-98-2	138	1.29